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# Network Analysis of Two-Stage Customer Decisions With Preference-Guided Market Segmentation

*Network-based analyses have effectively understood customer preferences through interactions between customers and products, particularly for tailored product design. However, research applying this analysis to diverse customers with varied preferences is limited. This paper introduces a market-segmented network modeling approach, guided by customer preference, to explore heterogeneity in customers' two-stage decision-making process: consideration-then-choice. In heterogeneous markets, customers with similar characteristics or purchasing similar products can exhibit different decision-making processes. Therefore, this method segments customers based on preferences rather than just characteristics, allowing for more accurate choice modeling. Using joint correspondence analysis, we identify associations between customer attributes and preferred products, characterizing market segments through clustering. We then build individual bipartite customer-product networks and apply the exponential random graph model to compare the product features influencing customer considerations and choices in various market segments. Using a US household vacuum cleaner survey, our method detected different customer preferences for the same product attribute at different decision-making stages. The market-segmentation model outperforms the non-segmented benchmark in prediction, highlighting its accuracy in predicting varied customer behaviors. This study underscores the vital role of preference-guided segmentation in product design, illustrating how understanding customer preferences at different decision stages can inform and refine design strategies, ensuring products align with diverse market needs. [DOI: 10.1115/1.4066420]*

**Keywords:** customer preference modeling, network-based analysis, market segmentation, customer decision-making process, data-driven engineering, model-based systems engineering

## 1 Introduction

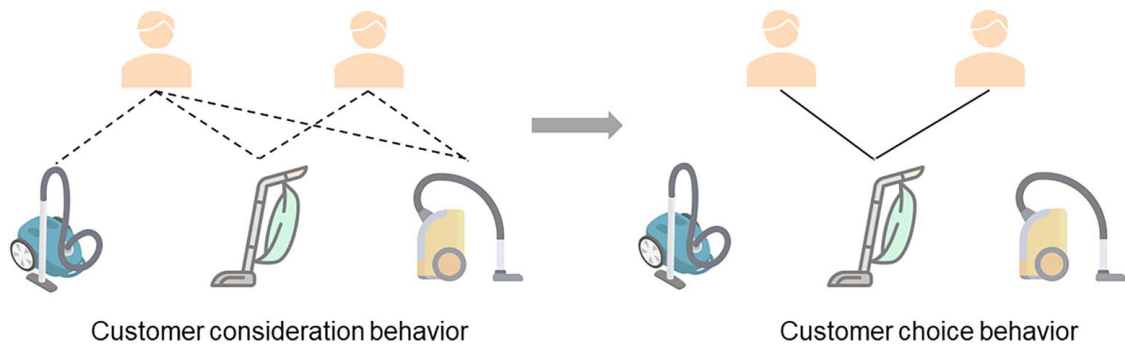
A quantitative understanding of customer preferences plays a vital role in product design, as it influences areas such as design attribute selection [1] and design optimization [2]. Network-based models have been increasingly used to quantitatively analyze customer preferences and behavior [3–5]. These models represent customers and products as nodes and their relations as edges in a network, allowing for the analysis of complex interactions and relationships between customers and products. One major advantage of

network-based models over traditional utility-based choice modeling in customer preference modeling [6] is their ability to handle both exogenous and endogenous attributes. Exogenous attributes can include product design features and customer attributes, while endogenous attributes may relate to the effects from the market structure [7].

Meanwhile, there is growing evidence in Refs. [8–10] on consumer research highlights a two-stage decision-making process in customers: consideration and choice. This process is exemplified in an illustrative process of vacuum cleaner purchase in Fig. 1. Initially, in the consideration stage, customers make initial selections of products to form a consideration set. Subsequently, in the choice stage, customers evaluate the tradeoffs among the products in the consideration set to make a final choice. The two-stage

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**Fig. 1 Two-stage consider-then-choose decision-making in an example of customers purchasing vacuum cleaners. The two-stage choice model assumes that each customer considers a subset of products first and then makes the final decisions.**

network-modeling approach [11] has shown effectiveness in more homogenous markets, such as the automotive sector, where customers buying the same category of products tend to have similar needs and preferences. In contrast, in diverse markets like vacuum cleaners, with customers choosing from a wide range of products, customers considering robotic vacuum cleaners might also evaluate the option of buying upright vacuum cleaners. In these markets, classifying customers solely by their purchased products is insufficient, as customers who make similar purchases still possess significantly distinct decision-making processes. Consequently, a systematic method for investigating heterogeneous customer preferences and segmenting customers by associating their heterogeneity with preferences is necessary for more diverse and mixed markets. Our *hypothesis* is that network modeling for individual market segments provides more accurate analysis than network modeling for the all-in-one market without segmentation.

In market-related research, segmentation categorizes customers into homogeneous groups with similar characteristics and needs [12,13]. While traditional segmentation, which considers factors like demographics [14], psychographic [14], behaviors [15], and needs [16], has proven effective in understanding customer preferences, its integration with network-based customer preference modeling for product design remains less explored. This paper aims to address this gap in the literature by developing market segmentation methods in the context of network-based modeling to enhance customer preference modeling and product design. Specifically, our research objectives include: (1) identifying market segmentation considering the association between customer heterogeneity and their preferences, (2) examining the impact of product attributes on the two-stage decision-making process through market-segmentation-based network modeling, and (3) evaluating the efficacy of network-based modeling that incorporates market segmentation in improving the prediction accuracy of customer considerations and choices.

To address our objectives, we propose a preference-guided market-segmentation-based network modeling approach, tailored to understand heterogeneous customer preferences in two-stage decision-making. In this approach, we first identify distinct customer preferences by categorizing different product communities based on customer considerations. Then we employ joint correspondence analysis (JCA) [17] to visualize the heterogeneity in customer preferences and their association with different customer attributes. Next, we cluster customers into different groups based on key customers' attributes and construct consideration and choice of bipartite networks for each group. Finally, the exponential random graph model (ERGM) [18], is utilized to investigate the key factors affecting customer consideration and choices in different market segments, which facilitate the product design process. A single network model without market segmentation serves as the baseline to be compared with our proposed model on their prediction accuracy. Our approach is demonstrated using the data from the household vacuum cleaner customer survey, which was

systematically designed to study multi-stage customer preference modeling [19]. The dataset contains 1011 customer observations of 267 variables, covering product attributes, purchase history, and customer attributes, offering a comprehensive view of the customer decision-making process within a diverse product market.

The remainder of the paper is organized as follows: Sec. 2 reviews relevant literature on network-based modeling of customer preferences and methodology. Section 3 outlines the systematic approach of preference-guided segmentation and two-stage network modeling process and its demonstration using the vacuum cleaner dataset. Finally, Sec. 4 explores our findings' implications in product design, addresses study limitations, and highlights the research contributions in Sec. 5.

## 2 Literature Review and Technical Background

In this section, we commence with an overview of emerging methodologies in **network-based customer preference modeling**, highlighting their efficacy in providing insight for product design. Additionally, we introduce the essential statistical tools utilized in this paper for modeling customer preferences.

**2.1 Network-Based Customer Preference Modeling.** Recent advances in customer preference modeling have significantly influenced product design, particularly through data-driven methods of extracting and analyzing customer needs and preferences. Studies by Jin et al. [20] and Taurob and Tucker [21] have underscored the importance of understanding customer needs and opinions for effective product design and development. Additionally, approaches proposed by Afshari et al. [22] and Jiang et al. [23] have contributed to optimizing product design under changing user preferences and evaluating product perceptual similarity. Natural language processing has also played a crucial role in design research, as highlighted by Siddharth et al. [24], by supporting the design process through techniques that analyze and interpret consumer opinions and other textual data. Amidst these developments, network-based methods have emerged as a more promising and comprehensive approach, offering nuanced insights in the modeling of customer preference.

The foundation of the network-based approach stems from [25] the framework of multidimensional customer-product networks. In this framework, customers and products are represented as nodes, with customer consideration and choice behaviors depicted as links that connect them. Furthermore, links within customer layers denote customer social relations among customers, while links within product layers indicate product association relations. This network representation of customer-product relations is a powerful tool in product design, as it captures the interdependence among customers' decisions by considering various relationships within the socio-technical system. Examples of such interdependence include the "peer effect," where customers' purchase decisions are influenced by their social circles, and "crowd effects"

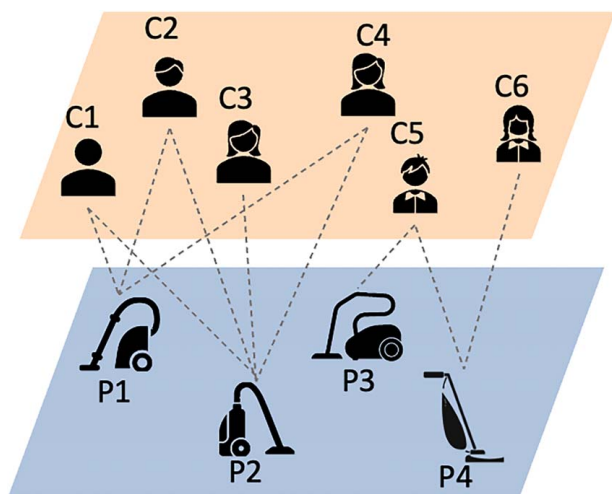
[26], in which products that gain more attention tend to attract a larger number of customer preferences. In contrast to traditional choice modeling methods that assume customers make decisions independently, network-based methods offer a more nuanced understanding of the actual decision-making process by accounting for these interdependencies. This understanding is crucial for product designers, as it enables them to design products that align with both individual preferences and the social context of their target customers.

In practice, acquiring data on customers' social relations and product associations can be challenging. However, in most product markets, it is feasible to collect data on customers' considered and purchased products, as well as customer and product attributes. This information can be utilized to create a bipartite network (as illustrated in Fig. 2), modeling the relations between customers and products. Although bipartite models cannot capture social influence among customers, they can still uncover the impacts of other network structures, such as market structure influence, on customers' behavior. For instance, these models can reveal whether a product's popularity increases customers' likelihood of purchase.

In the work of Fu et al. [11], a two-stage customer preference modeling study demonstrates that product attributes have different effects on customers' considerations and choices within the car market. Bi et al. [27] also employed bipartite network methods, focusing on analyzing multi-year data to capture the evolution of customers' preferences from the consideration stage to the choice stage over time. In the research conducted by Sha et al. [7], bipartite network models serve as choice modeling tools, where predicting customers' choices can be modeled as a link prediction problem. Although two-stage bipartite network models have been extensively explored, a significant research gap remains in the limited attention given to single product types in heterogeneous product markets, such as sedans in the car market. There is a need to investigate the integration of market segmentation methods with network-based analyses of diversified and mixed product markets. In such markets, products from different categories tend to be co-considered frequently, which necessitates a comprehensive examination of customer segmentation, offering valuable insights for more effective and nuanced product design strategies.

## 2.2 Technical Background

**2.2.1 Exponential Random Graph Model.** Recent advances in ERGMs have offered a unified and flexible statistical inference framework for network analysis [18]. ERGMs treat a network as



**Fig. 2** An example of a bipartite network: customers (C) and products (P) are represented by different types of nodes and their relationship (customers' consideration) is represented by links

a single entity, allowing researchers to model the interdependence (or network structure effects) among nodes [28]. For instance, in a star-shaped network with high centralization, nodes with numerous links tend to attract more nodes compared to those with few links. The key idea behind ERGMs is that they regard an observed network,  $y$ , as a specific instance from a set of potential random networks,  $Y$ , which follows the probability distribution in Eq. (1). These random networks all have the same set of nodes but vary in the configuration of their links. The randomization of these links can be customized to meet specific research objectives.

$$Pr(Y = y) = \frac{\exp(\theta^T g(y))}{\kappa(\theta)} \quad (1)$$

Here  $\theta$  is a vector of model parameters ( $\theta^T$  is the transpose of  $\theta$ ),  $g(y)$  is a vector of network statistics (including nodal attributes, edge attributes, and network structural effects), and  $\kappa(\theta)$  is a normalizing factor to ensure Eq. (1) is a proper probability distribution. Equation (1) suggests that the probability of observing any network is proportional to the exponent of a weighted combination of network characteristics: a particular network statistic  $g(y)$  is more likely to occur if the corresponding  $\theta$  is positive. For example, in a customer-product network, consider  $g(y)$  representing the number of connections between customers and the products that belong to the same product category. If the corresponding parameter  $\theta$  is positive, it is more likely for the network to have a high number of connections between customers and same-category products. Notably, in ERGMs, the entire network, rather than an individual link, is represented as the random variable  $Y$ , with its probability assessed based on the network configurations  $g(y)$ . ERGMs employ Markov Chain Monte Carlo simulation to estimate the parameter values  $\theta$  that maximize the likelihood of observed network structures [28].

Bipartite ERGMs are a type of ERGM specifically designed for modeling two-level network structures [29]. They follow the same model structure as defined in Eq. (1), except that  $g(y)$  must represent the network statistics specific to bipartite networks. Unlike one-mode networks, which consist of a single layer of nodes (e.g., social influence network and co-consideration network), bipartite networks are composed of links connecting two layers (two types of nodes). In this work, there are two types of links in two separate bipartite networks: consideration links and choice links.

The estimated parameters  $\theta$  in the bipartite ERGM can be used to infer the effects of product features and market structure (of corresponding  $g(y)$ ) on customer considerations and choices. By analyzing these effects, we can gain insights into the factors that drive customer decision-making and develop tailored marketing and engagement strategies to cater to the specific needs and preferences of each customer group.

**2.2.2 Joint Correspondence Analysis.** JCA is a statistical technique used to analyze and visualize the relationships between multiple categorical variables. Originating from correspondence analysis (CA)—a method for analyzing two-way contingency tables based on the singular value decomposition (SVD) of a matrix of a correspondence weight matrix—JCA extends CA to accommodate the analysis of multiple contingency tables or joint distributions [17]. This extension provides a more comprehensive view of the relationships by enabling the analysis of multiple variables simultaneously. Fundamentally, JCA is a dimension reduction method that enables visualization of a data matrix in a low-dimensional subspace.

Widely employed in customer preferences research, CA (or JCA) helps study the relationship between customers' preferences and their characteristics [30]. For example, CA was used to identify customer groups with similar purchasing patterns [31] and determine products that are most likely to be purchased together [30]. Additionally, the technique has been employed in social sciences to investigate the connection between demographics, attitudes, and behaviors [32]. In network-based customer preference modeling



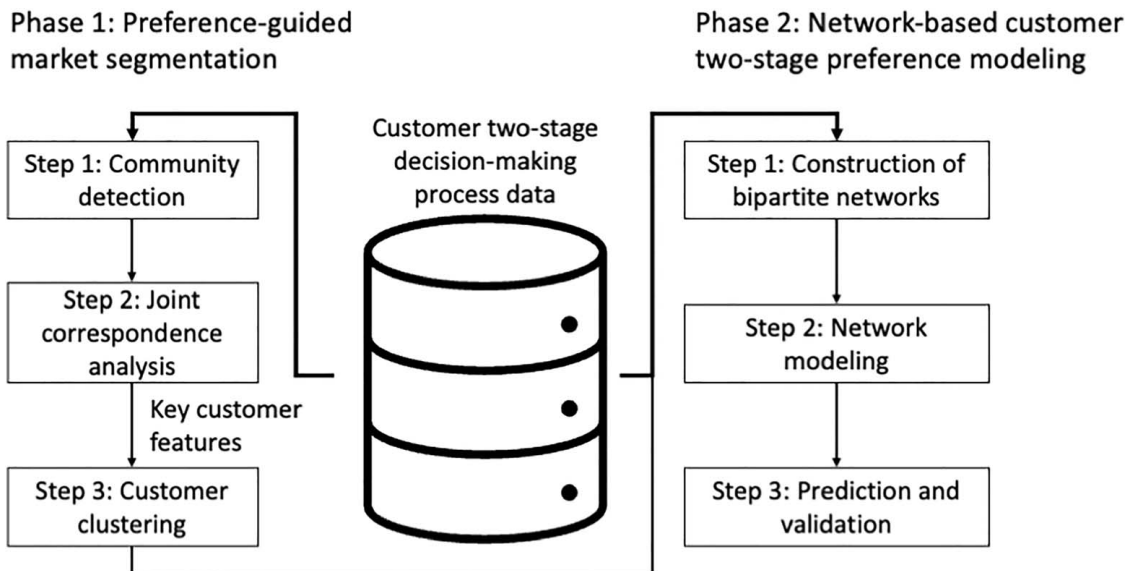


Fig. 3 Flowchart of the methodology and key components

[33], JCA was introduced as a multivariate approach that represents data graphically. This offers a visual understanding of the connections between product consideration sets and their relations with customer attributes.

### 3 Research Framework

In this section, we introduce the methodology along with a case study of the vacuum cleaner market.

**3.1 Data Description.** We demonstrate our methodology using customer survey data collected from the household vacuum cleaner market. The data contain **product attributes of vacuum cleaners** from major US online retailers. Additionally, our survey (detailed in Ref. [19]) has collected customer purchase histories and attributes. Through the Cint platform, we gathered 1011 valid responses from individuals who recently purchased vacuum cleaners. The survey was primarily focused on capturing the customers' two-stage decision-making process by collecting their **history of purchases**. The respondents reported the products they had considered and those they had ultimately purchased (with the option to specify up to six considered but not purchased products), and their considered and purchased products are taken as their consideration set. We also gathered comprehensive **customer attributes**, including demographic information, product usage contexts such as the frequency and location of vacuum cleaner use, and their general opinions about vacuum cleaners. The survey was thoughtfully constructed to avoid misinformation and to guarantee the integrity of the data collected.

In the proposed methodology, the customer attributes and their decision-making process are used to segment customers into different groups and construct the two-stage network, whereas the product attributes are used in the statistical modeling of the networks to investigate their effects on customers' decision-making.

**3.2 Overview of Methodology.** The methodology mainly contains two phases: (1) preference-guided customer segmentation informed by product community detection and joint correspondence analysis, and (2) network-based customer two-stage preference modeling. Figure 3 presents a flowchart detailing the main components within each phase.

### 3.3 Phase 1: Preference-Guided Customer Segmentation.

In the initial phase, we propose a preference-guided customer segmentation method. We first detect different types of customer preferences by utilizing product community detection methods. This involves identifying the products that customers frequently consider together. Then, we employ JCA as an exploration tool to examine the relationship between customer characteristics and their preferences. These preferences are indicated by the products they consider. JCA facilitates in identifying the key customer attributes that drive their preferences, which are then utilized for clustering customers into distinct segments.

**3.3.1 Step 1: Product Community Detection.** First, we identify different types of customer preferences, represented by the product communities to which their considered products belong. In this context, product communities refer to groups of products that customers frequently co-consider. To identify product communities, we first use their co-consideration relationships that reported in the survey data to construct a product association network, which presents an aggregated level of customer preferences on the whole market.

In network theory, a community refers to a group of nodes within a network that are highly interconnected with one another, but relatively sparsely connected to nodes outside the community. Community detection algorithms partition a network into such groups. By applying these algorithms to a product co-consideration network (as illustrated in the network diagram in Table 1), we can identify groups of products that are frequently co-considered. In the co-consideration network, each node represents a product, and the links between nodes indicate that the products are co-considered by at least one customer. The size of each node is proportional to its degree, representing the number of co-considerations associated with the product.

In our case study, the co-consideration network analysis reveals the presence of isolated nodes that are either not co-considered by other customers or only co-considered within 2 to 3 products. To ensure meaningful analysis, only the largest connected component in the co-consideration network is retained. This step removes 6% of the products and leaves 528 products for community detection.

The community detection algorithm we employ is based on a modularity-based optimization approach, as proposed by Newman and Girvan [34]. Modularity score quantifies the extent of connectivity within communities relative to what would be expected in a

**Table 1 Product community detection and characteristics (mean value and (standard deviation)) of each community**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Comm. | Descriptions               | Dominant type | Suction power (rating) | Price (dollar)  | Weight (lb.) | Representative models                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------|---------------|------------------------|-----------------|--------------|-------------------------------------------------------------------------------------|
| <br> Community 1<br> Community 2<br> Community 3<br> Community 4 | 1     | High-tech and expensive    | Robotic       | 2.62 (1.25)            | 296.12 (205.22) | 8.73 (5.09)  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2     | Traditional and affordable | Upright       | 2.90 (1.20)            | 179.37 (166.19) | 11.68 (6.36) |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3     | Strong suction power       | Upright       | 2.99 (1.34)            | 259.85 (218.60) | 10.20 (6.78) |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4     | Innovative and portable    | Stick         | 2.75 (1.22)            | 228.47 (201.72) | 8.77 (5.76)  |  |

**Table 2 Indicator matrix in joint correspondence analysis, with observations of customers' considerations as row entries, and considered product and customer attributes as column entries**

|            | Considered product community $x_1$ |        |        | Income $x_2$ |     |      | Pet $x_3$ |    | Quality is important $x_4$ |         |          |
|------------|------------------------------------|--------|--------|--------------|-----|------|-----------|----|----------------------------|---------|----------|
|            | Com. 1                             | Com. 2 | Com. 3 | Low          | Mid | High | Yes       | No | Agree                      | Neutral | Disagree |
| Customer 1 | 1                                  | 0      | 0      | 0            | 0   | 1    | 1         | 0  | 1                          | 0       | 0        |
| Customer 1 | 0                                  | 1      | 0      | 0            | 0   | 1    | 1         | 0  | 1                          | 0       | 0        |
| Customer 2 | 1                                  | 0      | 0      | 0            | 1   | 0    | 0         | 1  | 0                          | 0       | 1        |
| Customer 2 | 0                                  | 1      | 0      | 0            | 1   | 0    | 0         | 1  | 0                          | 0       | 1        |
| Customer 2 | 0                                  | 1      | 0      | 0            | 1   | 0    | 0         | 1  | 0                          | 0       | 1        |
| Customer 3 | 0                                  | 0      | 1      | 0            | 0   | 1    | 0         | 1  | 0                          | 1       | 0        |

random network. Many algorithms automatically detect the number of communities by optimizing the modularity score. However, in less dense networks, this approach may return a large number of communities, making it challenging to derive insights from the result of community detection. To address this, we use the elbow rule [35] to pre-define an appropriate number of communities based on a reasonably high modularity score.

In the product community detection process, we apply the Spinglass algorithm [36], which is a flexible approach that allows users to pre-define the number of communities and provides superior modularity scores compared to other algorithms with similar functionality, alongside the elbow rule to determine the optimum community size is four.

In Table 1, we provide the community detection result and summarize the key characteristic with the mean value and standard deviation of each community.<sup>2</sup> On average, community 1 contains the most expensive vacuum cleaners, and the most dominant type is robotic vacuum cleaners, which is summarized as “high-tech and expensive.” Community 2 contains the vacuum cleaners with the lowest price and the most dominant type is upright vacuum cleaners, which can be named as “traditional and affordable.” In community 3, the highest suction power is observed, which is represented by the attribute “strong suction power.” Finally, community 4 is dominated by lighter-weight stick vacuums, which are described

as “innovative and portable.” We represent customers' preferences by the product communities of the products they have considered. This representation is effective because each product community is composed of closely interconnected products, which are often considered together by customers with similar tastes. Consequently, the specific product community associated with a customer's considered products can provide a general indication of that customer's preferences.

The identified product communities represent the groups of products that are frequently considered together by customers, indicating a certain similarity in customer preferences within each community. Instead of using standard product categories like upright, stick, and robotic vacuum cleaners for product differentiation, we have chosen product communities. This approach is aimed at more accurately capturing customer preferences, as these communities more closely reflect consumer perceptions and categorizations of products.

**3.3.2 Step 2: Joint Correspondence Analysis.** After detecting product communities in the product co-consideration network, we employ JCA to explore the association between customer characteristics and their preferences for vacuum cleaners, which was represented by the product community that their preferred product belongs to. We consider a range of customer attributes, including demographic attributes, usage context attributes, and personal viewpoints about products. When using JCA, we consider  $N$  customers' consideration observations, each associated with  $x_1, \dots, x_q$

<sup>2</sup>All vacuum cleaner images are sourced from their respective official websites.

categorical variables, which include both customer attributes and the product communities they have considered. Each  $x_j$  has  $L_j$  number of levels ( $j \in [1, q]$ ). From this data, we can generate a binary indicator matrix  $\mathbf{Z}^{(j)}$  of dimension  $N \times L_j$  for each categorical variable  $x_j$ , such that  $Z_{il}^{(j)} = 1$  if and only if  $x_{ij} = l$ , where  $l$  is a level of  $x_j$  ( $i \in [1, N]$ ). These individual matrices  $\mathbf{Z}^{(j)}$  can then be concatenated to form a large indicator matrix  $\mathbf{Z}$  with dimensions  $N \times J$ , where  $J = L_1 + \dots + L_q$  represents the total number of categorical levels for all input variables  $x$ . Table 2 displays an example of the indicator matrix  $\mathbf{Z}$  with three customers (with six customer consideration observations) in rows and four categorical variables in columns, including the product community variable  $x_1$  that list the product community for customers to consider. Because the indicator matrix  $\mathbf{Z}$  might consume significant memory resources when dealing with a large number of respondents and categorical levels, JCA operates on the Burt matrix  $\mathbf{B} = \mathbf{Z}'\mathbf{Z}$ , defined as the cross-tabulation of all categorical levels. Using the Burt matrix  $\mathbf{B}$ , the column coordinates relative to the principal axis can be calculated through SVD, and JCA's corrected inertia is obtained by iteratively updating the solution.

JCA transforms high-dimensional datasets into low-dimensional spaces. By projecting variable levels into this space, a JCA plot reveals relationships between customer attributes and considered product communities. In the case study, we evaluated 31 customer characteristics related to demographic attributes, usage contexts, and personal opinions about vacuum cleaners, which comprised a total of 152 distinct category levels. These levels and product communities are depicted in Fig. 4. Specifically, four separate product communities are denoted in different circles. Important feature levels are represented as black dots, while less important ones are indicated by gray dots. Feature importance is determined by considering the **proximity** to the four product communities and the value of a **feature level's inertia** [17].

**Proximity between points** In the plot, each customer attribute level and their considered product communities are represented by a point. Closely positioned points indicate a stronger association between the corresponding categories (either customer attributes or product communities). When two product communities are proximal, it suggests that they are often considered by customers with similar profiles. Two customer attributes are closer if they often appear together for specific customers. Last, when a product community and customer attributes are located near each other, it implies that customers who consider products in that community frequently possess those attributes. In our analysis, we focus on identifying customer attributes that are closer to each product

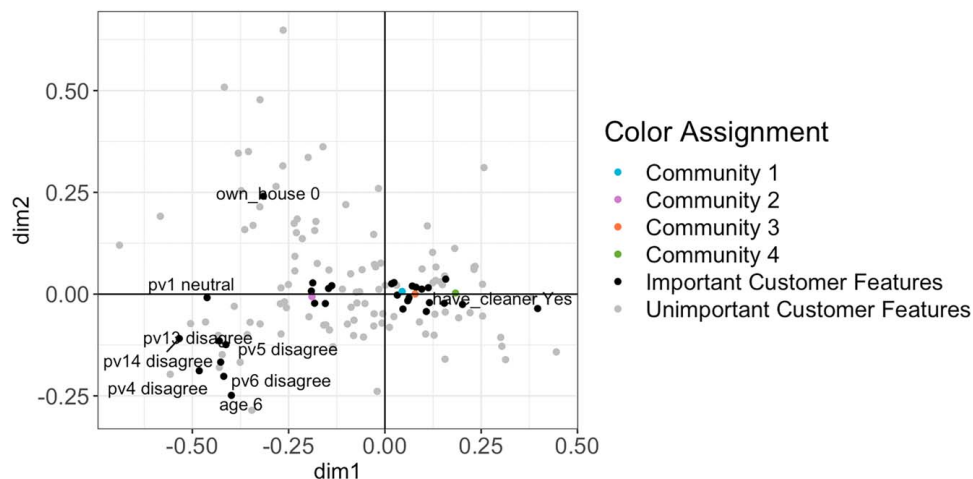
community. This indicates a strong preference for certain products among customers with those attributes.

The proximity between each product community and customer feature levels can be observed from the enlarged JCA plots for each community (Fig. 5). The closer a product community to a customer feature on the plot, the more frequently customers with the specific feature tend to consider products from that community.

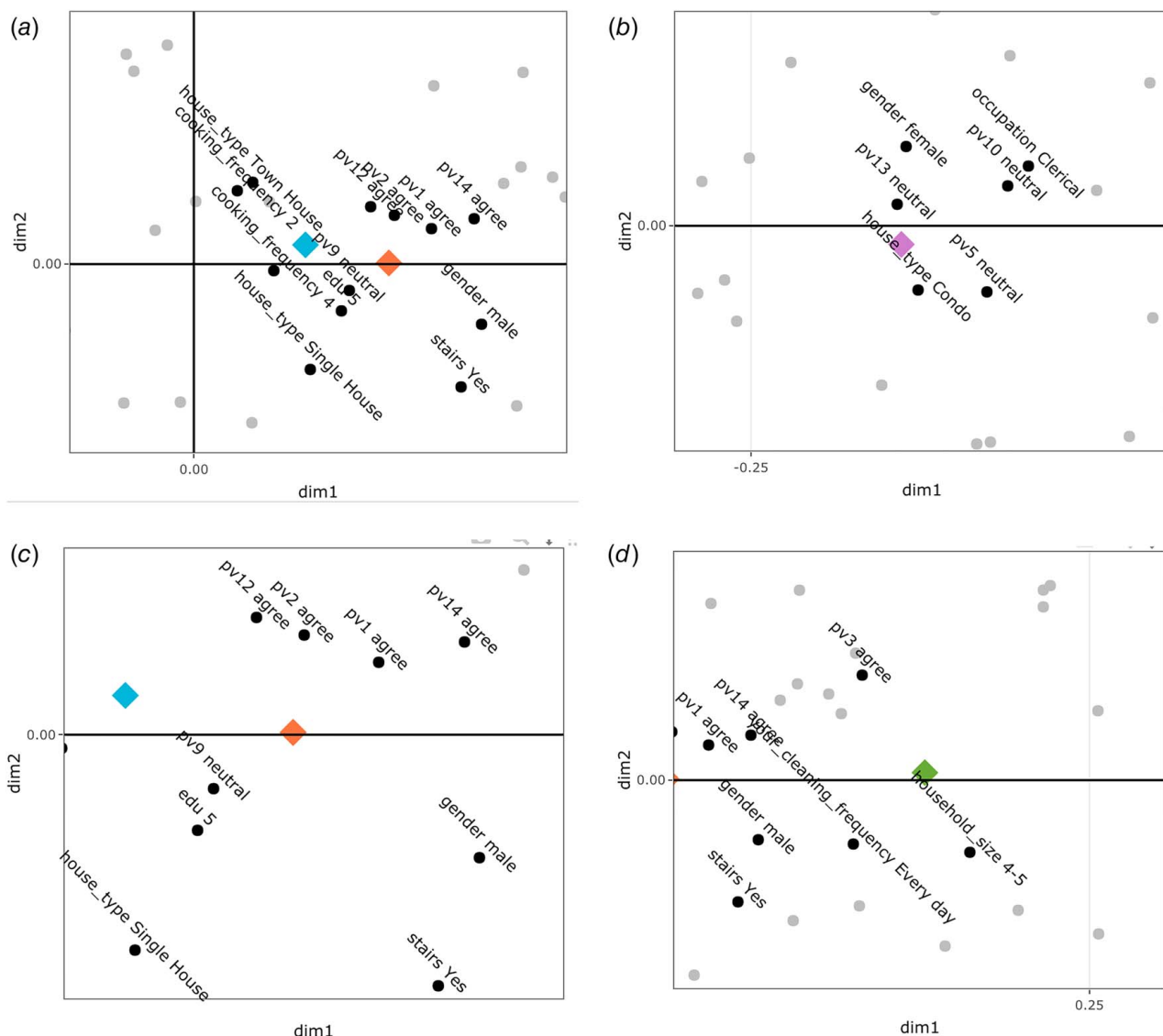
For product community 1 (high-tech and expensive), it is strongly associated with customer features such as house type (townhouse, single house), cooking frequency (3 to 4 days a week, every day), bachelor's degree (Edu 5), tech enthusiasts (PV2 agree), and impressionable (PV12 agree). Product community 2 (traditional and affordable) is highly associated with customer features including occupation (clerical), gender (female), house type (condo), neutral opinions on styling (PV5), after-sale service (PV10), and advocacy (PV13). Product community 3 (strong suction power) is closely related to house type (single house), stairs (yes), gender (male), innovation enthusiasts (PV1 agree), tech enthusiasts (PV2 agree), impressionable customers (PV12 agree), and quality-conscious customers (PV14 agree). Finally, product community 4 (innovative and portable) is strongly associated with features such as cleaning frequency (every day), house size (4–5), lifestyle-conscious customers (PV3 agree), and brand advocates (PV13 agree).

**Column inertia** Column inertia is a measure of the variation or dispersion of category levels in the multidimensional space. It quantifies the “distance” between each category level and the average or centroid for the respective category in the space defined by JCA. In simpler terms, it conveys how different or distinct a category level is compared to the average of all category levels in the same category. Higher inertia values signify that a category level is more distinct from the average, while lower values indicate its proximity to the average. In our case, we believe that a higher column inertia value suggests that a customer attribute level is more diverse or different from the average, indicating the potential significance in shaping customer preferences of the corresponding attributes.

In Fig. 4, these high-inertia features are labeled with level names. Based on the JCA results, the top ten features with the highest inertia values were selected, which include: PV5 (disagree with the importance of styling), PV4 (disagree with the environmentally friendly feature), Occupation (Retired), Own house (0), PV14 (disagree with the highest quality being important), Age (6), PV13 (disagree with advocating for a favorite brand), PV1 (neutral toward innovation), PV6 (disagree with the importance of energy efficiency), and Have house cleaner (Yes).



**Fig. 4 Joint correspondence analysis based on vacuum cleaner communities and customer attributes. The feature levels with high inertia are labeled with text. Product community 2 falls in the region of  $\text{dim1} < 0$ , and product communities 1, 3, and 4, fall in the region of  $\text{dim1} > 0$ .**



**Fig. 5** Customer features that have a strong association with each product community (proximity): (a) the closest customer features to community 1, (b) the closest customer features to community 2, (c) the closest customer features to community 3, and (d) the closest customer features to community 4

Besides, we consider other results from the JCA analysis. **Explained variance in the low-dimensional space:** the percentage of the total variation explained by the first two dimensions represents how well they can represent the data variation. **Opposite quadrants:** product communities in opposite quadrants of the plot tend to have negative relationships, meaning that they are preferred by different customer segments or have contrasting associations with customer attributes. For example, it is evident that communities 2 and 4 are more distinguishable than communities 1 and 3. Also, community 2 is located in the opposite quadrant to communities 1, 3, and 4. This observation suggests that customers with a strong preference for community 2 products have distinct preferences compared to those who favor products in communities 1, 3, and 4. Additionally, the first two dimensions account for 60% of the data variance, indicating that these dimensions have effectively captured a substantial amount of information present in the raw customer dataset.

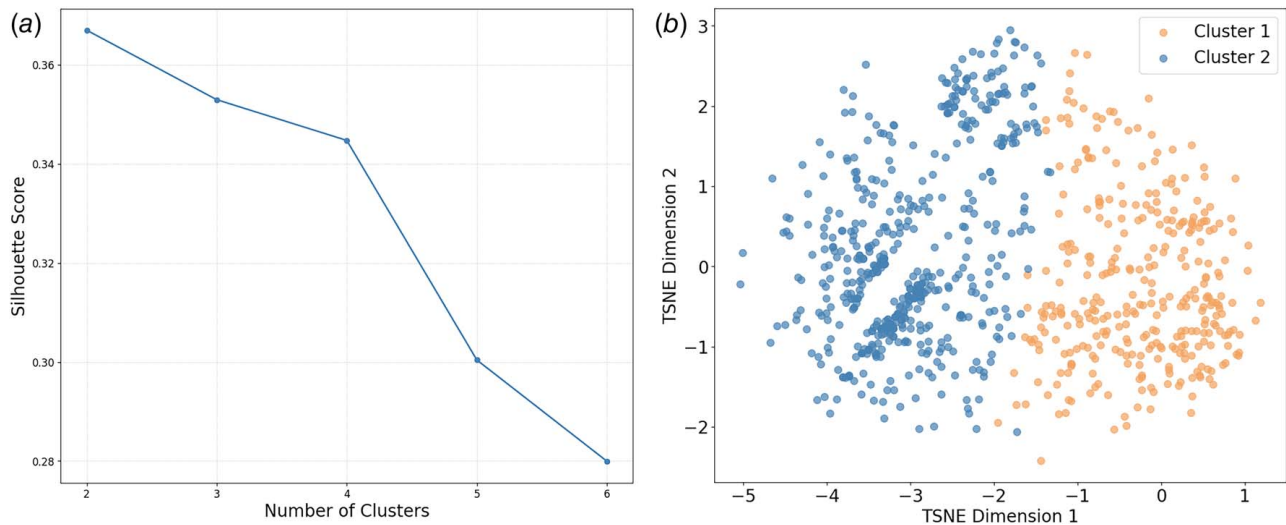
In summary, the JCA plot shows that there are at least two major categories of customers, separated by  $\text{dim1} > 0$  and  $\text{dim1} < 0$ , respectively. Their preferences significantly differ as they opt for vacuum cleaners associated with distinctive product communities. Utilizing the JCA results (proximity and column inertia), we have

identified candidate features related to customer preferences that can be employed for effective customer segmentation.

**3.3.3 Step 3: Customer Clustering.** Using key features associated with customer preferences identified through JCA analysis, we can categorize customers into distinct groups with unsupervised clustering. The customer clustering process includes selecting an appropriate clustering algorithm, determining the optimal number of clusters, and validating the clustering results with visualization and statistical tests, which evaluate differences between customer groups.

Drawing upon the JCA results, the analysis utilizes a comprehensive set of 20 features spanning demographic attributes, usage context attributes, and personal viewpoints attributes for customer segmentation. The selected customer attributes are all categorical, which can be further classified into nominal variables (e.g., gender and house type) and ordinal variables (e.g., education level, household size, and personal viewpoints). We use k-prototype clustering algorithm [37] to cluster the customers using these attributes. The algorithm effectively handles mixed data types by combining the k-means and k-modes algorithms. In this particular case, the Hamming distance is employed for nominal variables, while the Euclidean distance is used for ordinal





**Fig. 6 Customer segmentation results: (a) silhouette score analysis for optimal cluster count: Two clusters yield the best results and (b) t-SNE visualization of k-prototype clustering results in two dimensions**

variables. This approach, which takes into account the inherent order present in ordinal variables, yields better results in terms of silhouette score than the one that ignores this order.

To determine the optimal number of clusters, a silhouette score analysis is conducted. As depicted in Fig. 6(a), the silhouette score varies with the number of clusters, and the highest score is obtained with two clusters. This suggests that the clustering solution with two clusters maximizes the similarity within clusters while maximizing the dissimilarity across clusters. Using the k-prototype algorithm, we obtain 331 customers in cluster 1 and 540 customers in cluster 2. The clustering results are visualized with a t-SNE plot, which reduces the customers' attributes to a low-dimensional space and reflects their similarity through the proximity of the points on the map, as shown in Fig. 6(b).

This comprehensive segmentation methodology is designed to provide a deeper understanding of customer heterogeneity, enabling the development of tailored marketing and engagement strategies for each identified customer group. By addressing the unique needs and preferences of each group, we anticipate more accurate and effective results in subsequent stages of the study.

**3.4 Phase 2: Two-Stage Customer Decision-Making Process Network Modeling.** In the second part of this study, bipartite customer-product networks are constructed, modeled, and interpreted for the different customer groups. These networks aim to reveal the underlying patterns of customer preferences among different groups and their interactions with diverse products. Simultaneously, the network-based model can uncover the underlying market structures. To further validate the model, a portion of the customers' considerations and choices is withheld during the modeling process. We calculate our model's accuracy on these withheld preferences and compare it to that of a benchmark model that doesn't take market segmentation into account.

**3.4.1 Step 1: Construction of Consideration and Choice Networks.** When representing customer-product relationships with bipartite networks, customers and products are modeled as two distinct types of nodes. The customer decision-making process can be modeled as a probability of forming consideration or choice links between the two types of nodes, and the customers' choices are made among the products being considered.

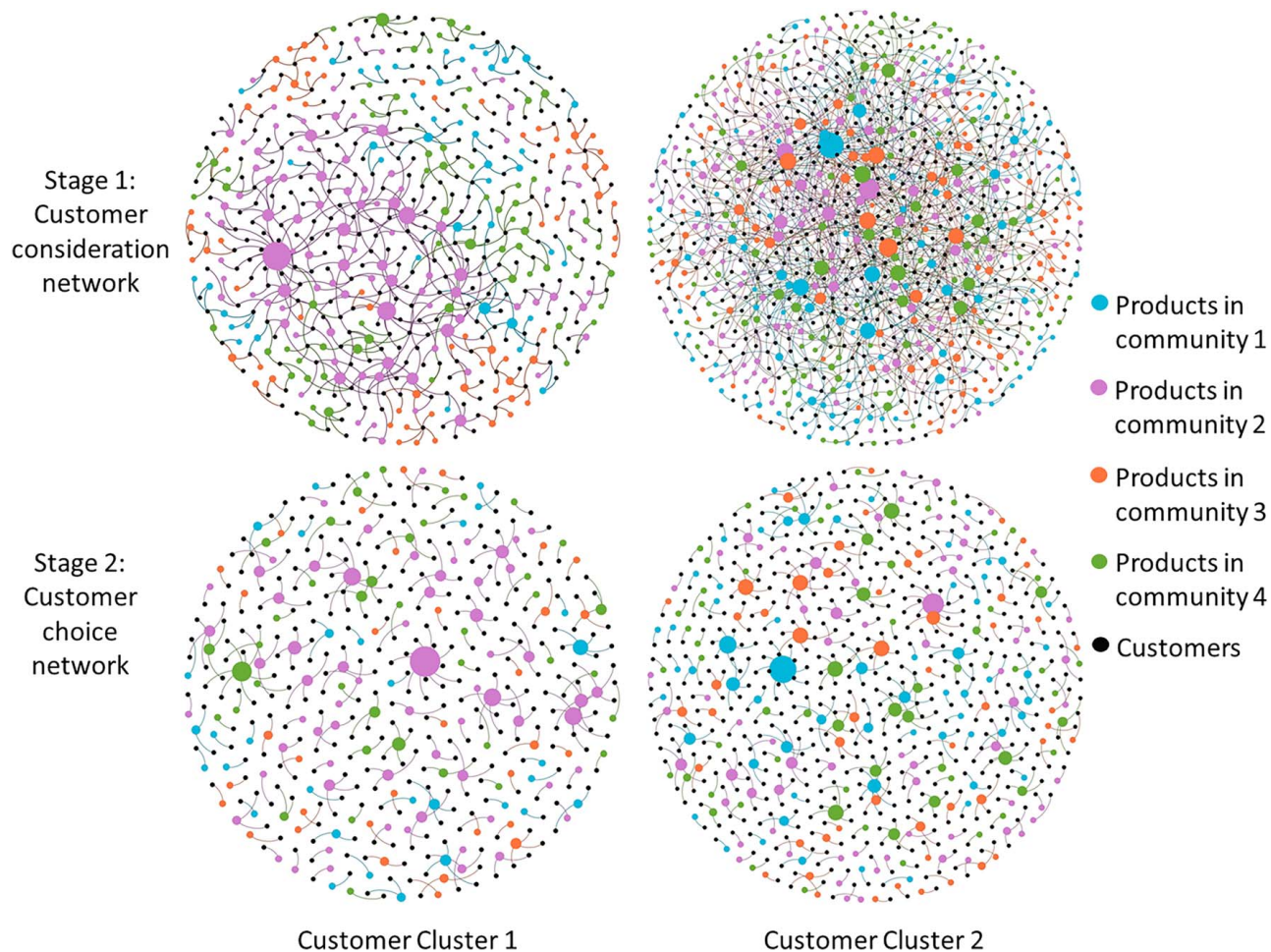
After removing customers who have only considered and purchased vacuum cleaners that are isolated in the co-consideration network, there are 871 vacuum cleaner customers that are classified

into two market segments. We construct separate bipartite networks to analyze each market segment's customer preferences toward product attributes. For validation purposes, 10% of customers are reserved as testing datasets, meaning that their consideration and purchase information, which are represented as links in the constructed networks, are unknown. The customer-product bipartite networks at the consideration stage and purchase stage are plotted in Fig. 7, where black dots are customers and colored dots are products associated with different communities. Even though customers may consider multiple products, they only purchase one from the consideration set. That is why the network for stage 1 (consideration) is much denser than that for stage 2 (choice), and the links in the choice network are conditional on the consideration stage. The size of each vacuum cleaner node reflects its level of popularity, which is determined by the frequency of its consideration or selection within the network. In both the consideration stage and the choice stage, the products from community 2 (traditional and affordable) are more prevalent among customers in cluster 1 (price sensitive).

The networks constructed for customer clusters are characterized as follows: cluster 1 (price sensitive)'s network comprises 331 customers who considered 286 vacuum cleaner models, with a consideration network density of 0.0031 and a choice network density of 0.0016. Network density, in this context, represents the proportion of existing connections between nodes (customers and products) relative to the total possible connections in the network. In comparison, cluster 2 (innovation passionate)'s network includes 540 customers who evaluated 467 models, with a consideration network density of 0.0025 and a choice network density of 0.0010. The choice networks are sparser than the consideration networks as each customer considers several products while only purchasing one. The data reveal that the two networks have comparable densities, although the network for cluster 2 exhibits a larger size in comparison to the network for cluster 1. Additionally, significant overlaps were observed within each cluster, where different customers considered the same models, indicating shared preferences among individuals. For example, the network visualization distinctly reveals that product community 2, characterized by its traditional and affordable aspects, predominates among customer cluster 1, which is notably price sensitive.

**3.4.2 Step 2: Statistical Network Modeling.** For the constructed consideration and purchase networks, we use the ERGM formula in Eq. (1) to build the statistical model. As introduced in Sec. 2, ERGM takes the observed network  $Y$  as the dependent variable, which is the consideration network and purchased network,





**Fig. 7 Bipartite consideration and choice networks in different customer clusters. Products in community 2 are more prevalent in customer cluster 1.**

respectively in stage 1 and stage 2. The input variables are network statistics  $g(y)$ , and they comprise three different levels: **Network structure effects**, which assess the prevalence of specific network structures and test the hypothesis about common market structures studied in economics. For example, we can test whether a market is oligopoly, where a few companies dominate the entire market, or competitive, where several companies vie for market share.

**Nodal attributes**, which measure the influence of product attributes on customers' consideration and choice stages. **Homophily effects**, refer to the tendency of customers to consider products with similar attributes.

The descriptive analysis of the networks is summarized in the last section and the descriptive analysis of the product features used in the ERGM is summarized in Table 3.

**Table 3 Descriptive analysis of product features in ERGMs**

| Feature types | Features             | Cluster 1       |                  | Cluster 2       |               |
|---------------|----------------------|-----------------|------------------|-----------------|---------------|
|               |                      | Mean (std)      | Range            | Mean (std)      | Range         |
| Numerical     | Price (\$)           | 232.79 (177.90) | [23.44, 1209.99] | 242.15 (203.41) | [19.99, 1499] |
|               | Capacity (L)         | 1.52 (3.34)     | [0, 53]          | 1.89 (4.22)     | [0, 53]       |
|               | Suction power rating | 2.81 (1.24)     | [1, 5]           | 2.82 (1.26)     | [1, 5]        |
|               | Value                | Percentage      |                  | Value           | Percentage    |
|               | Upright              | 1               | 43.01%           | 1               | 30.13%        |
|               |                      | 0               | 56.99%           | 0               | 69.87%        |
| Categorical   | Robotic              | 1               | 15.73%           | 1               | 20.73%        |
|               |                      | 0               | 84.27%           | 0               | 79.27%        |
|               | Filter type          | HEPA            | 41.26%           | HEPA            | 43.38%        |
|               |                      | Non-HEPA        | 58.74%           | Non-HEPA        | 56.62%        |
|               | Bag                  | Bagged          | 8.39%            | Bagged          | 11.75%        |
|               |                      | Bagless         | 91.61%           | Bagless         | 88.25%        |
|               | Surface type         | Multi Surface   | 60.14%           | Multi Surface   | 55.34%        |
|               |                      | Hard Floor      | 5.24%            | Hard Floor      | 7.48%         |
|               |                      | Carpet          | 5.24%            | Carpet          | 5.34%         |
|               |                      | All floors      | 29.37%           | All floors      | 31.84%        |

**Table 4 ERGM results of two-stage modeling results for different customer clusters (market segmentation)**

| Model terms             | Cluster 1: Price sensitive |            | Cluster 2: Innovation passionate |            |
|-------------------------|----------------------------|------------|----------------------------------|------------|
|                         | Stage 1                    | Stage 2    | Stage 1                          | Stage 2    |
| Edges                   | −6.0440 ***                | 1.0120.    | −5.7270 ***                      | −0.7479 *  |
| Market distribution     | 1.1770 ***                 | −0.7747 ** | −0.4156 *                        | −0.3547 *  |
| Product-shared partners | −0.3785 ***                |            | −0.3745 ***                      |            |
| Upright vacuum          | 0.7580 ***                 | −0.0763    | 0.2183 ***                       | 0.6046 *** |
| Robotic vacuum          | 0.3343.                    | −0.0824    | 0.4267 ***                       | 0.0244     |
| Price                   | −0.0008.                   | 0.0001     | 0.0004 **                        | 0.0002     |
| Filter: HEPA            | −0.3595 **                 | 0.2484     | −0.1533 *                        | 0.0612     |
| Capacity                | 0.0256                     | −0.0363    | 0.0231 ***                       | 0.0273 *   |
| Bagless                 | 0.5097 *                   | −0.1515    | 0.4910 ***                       | 0.3286     |
| Suction power           | 0.1264.                    | −0.0919    | 0.1620 ***                       | −0.0771    |
| Surface match           | 0.2108 **                  |            | 0.2623 ***                       |            |
| Model fit: AIC          | 6997                       | 1427       | 15,268                           | 2609       |
| Model fit: BIC          | 7109                       | 1511       | 15,392                           | 2712       |

Note:  $p < 0.1$ ; \*  $p < 0.05$ ; \*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$ .

Specifically for modeling the two-stage decision-making process using the ERGM framework, we implement a constraint in the randomization process to ensure that only product nodes connected during the consideration stage can be linked to customers in the choice stage by setting the links outside these considerations impossible [7]. By doing so, we accurately represent the process where customers choose products exclusively within their consideration sets.

Table 4 presents the estimated parameters (with significance levels) of the ERGM used in the two-stage decision-making process for two distinct customer clusters.

The **network structural effects** include three elements in our model: “edges,” “market distribution,” and “product shared partner.” The edges measure the number of edges in a network and serve as a baseline characteristic. Figure 8 provides a graphical illustration of the other two network structure effects and their interpretation in customer-product networks. The market distribution<sup>1</sup> is measured with an endogenous structural variable of ERGM—geometrically weighted degree distribution (gwb2degree) [18]. This variable characterizes the distribution of the degrees among the vacuum cleaner models. A positive coefficient for market distribution indicates a more even distribution among degrees (i.e., most vacuum cleaners have similar sales, as depicted in Fig. 8(a)), while a negative coefficient implies a skewed distribution (i.e., a few vacuum cleaners have significantly higher sales than others as illustrated in Fig. 8(b)). Notably, for customer cluster 1 at the consideration stage, the market distribution exhibits a positive and significant effect, implying that most vacuum cleaners have comparable chances of being considered by customers. In contrast, for customer cluster 2, the market is more skewed in the consideration stage. This trend is further evidenced by the product degree distribution plot for the consideration stage shown in Fig. 9. In this plot, product nodes within cluster 2 exhibit a long-tail effect, indicating that a small number of products receive significantly more attention compared to the rest. However, in the choice stage, both networks in cluster 1 and cluster 2 are skewed, indicating that final sales are more concentrated toward a few products in the final choice.

The “product shared partner” is measured by geometrically weighted dyadwise shared partner distribution (gwb2dsp) [18], which is specifically relevant to consideration networks. This metric helps us understand how pairs of product nodes that have shared connections to some customers tend to form connections with other customer nodes. In other words, it helps us see how products that are popular with the same customers might be related to each other. Intuitively, if two products are co-considered by a customer, they are more likely to be co-considered by additional customers, suggesting that customers tend to consider the same products. However, the model results indicate negative coefficients for product shared partners, revealing that products are often co-considered by a few

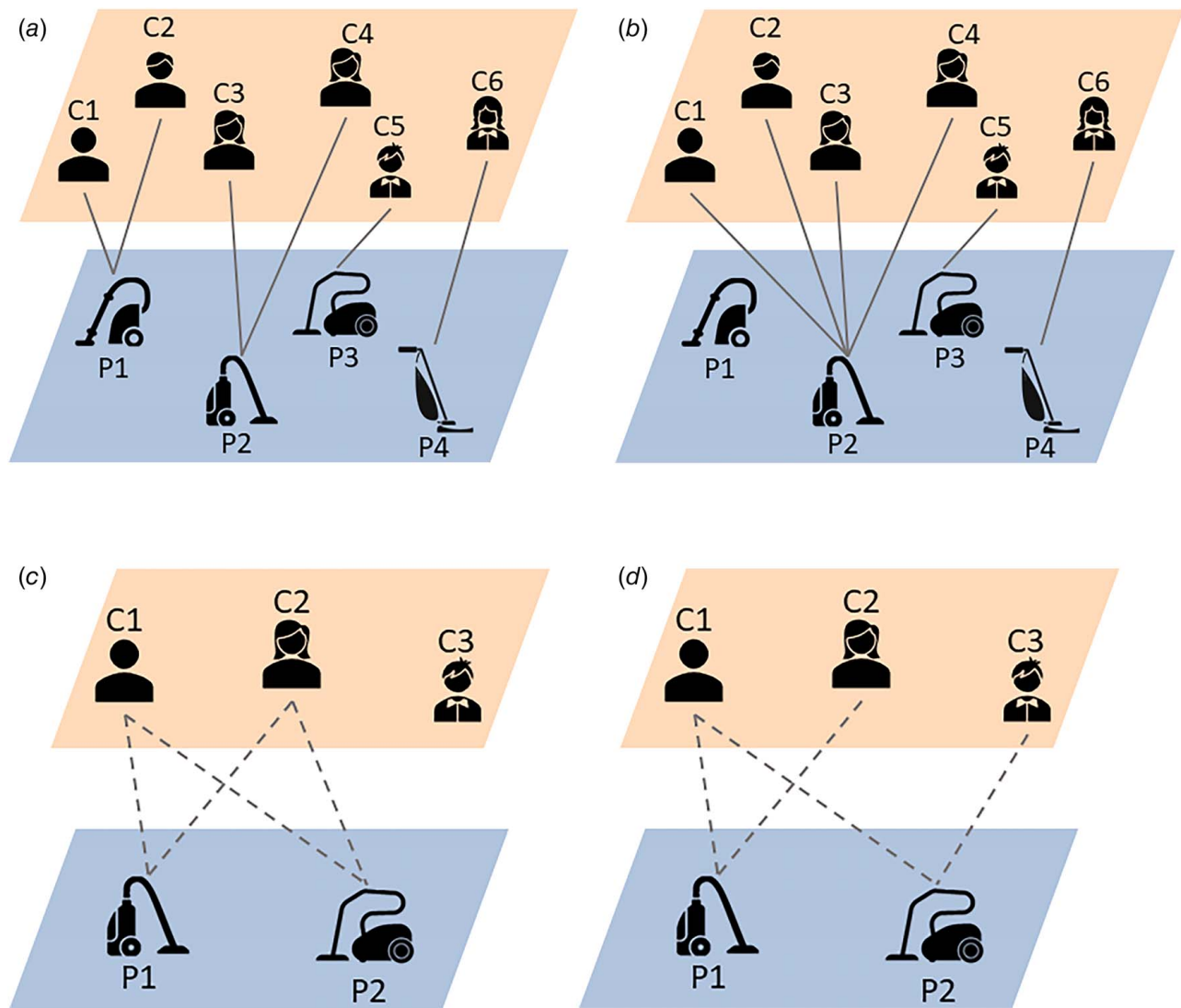
people (as illustrated in Fig. 8(d)). This observation can be attributed to the limited size of the customer sample in the survey data, which results in a lack of shared considerations among customers. Additionally, the diverse nature of the vacuum cleaner market may contribute to this phenomenon, as a wide variety of available products prevents customer considerations from concentrating on just a few options.

The estimation of **nodal attributes** highlights the significance of vacuum cleaner product attributes in the customer consideration and choice stages. A positive effect for nodal attributes indicates that a product with these attributes is more likely to be considered or chosen. In the consideration stage, both upright and robotic vacuum cleaners are popular for both customer clusters. However, innovation-passionate customers (cluster 2) demonstrate a much stronger preference for robotic vacuum cleaners, while price-sensitive customers (cluster 1) favor upright vacuum cleaners. In terms of “price,” customer cluster 1 is more price sensitive, as the price has a negative effect in their consideration stage, while customer cluster 2 demonstrates lower price sensitivity. Regarding engineering attributes, it appears that vacuum cleaners with “HEPA filters” (more efficient filters) do not significantly increase customers’ interest in the consideration stage. Although both customer clusters favor “bagless” and strong “suction power” products, customer cluster 2 exhibits a stronger preference compared to cluster 1. Furthermore, they also show a keen interest in vacuum cleaners with larger “capacities.”

We also observe that in the choice stage, customer cluster 2 finds upright vacuum cleaners more appealing. This implies that despite a strong interest in innovation-passionate customers to consider robotic vacuum cleaners, they often end up selecting upright vacuum cleaners from their consideration set. This preference is supported by the fact that among all survey respondents, 47.03% of them selected upright vacuum cleaners, while the remaining 52.97% chose from robotic, stick, handheld, and canister vacuum cleaners. Meanwhile, product features are less influential in customer decision-making during the choice stage compared to the consideration stage. Only “capacity” has marginally significant effects on customer cluster 2. This may be because the model cannot take into account other important factors, such as customer ratings and online recommendations, which can influence the choice stage. Additionally, most product features have already been evaluated during the earlier consideration stage.

The network models effectively capture **homophily effects**, which are analyzed during the consideration stage to determine if product similarity influences customers to consider them together. By incorporating the model term “surface match,” we observe significant and positive impacts of the feature, indicating that customers are more likely to consider products with the same “surface recommendation” (e.g., hardwood, carpet).

By comparing the key factors in the customer decision-making process across multiple stages, we can discern the different



**Fig. 8** Illustration of network structural effects in customer-product networks with distinct layers for customers and products: (a) positive market distribution effect demonstrates a balanced market distribution, with customers' considerations or choices evenly distributed, leading to products receiving similar levels of attention, (b) negative market distribution effect depicts a skewed market distribution, where customers' considerations or choices are less evenly distributed, resulting in a few products gaining greater popularity, (c) positive product shared partner effect highlights the tendency for pairs of products (P1 and P2) that are co-considered by one customer to be more likely co-considered by other customers as well, and (d) negative product shared partner effect illustrates the scenario where pairs of products (P1 and P2) that are co-considered by one customer become less likely to be co-considered by other customers

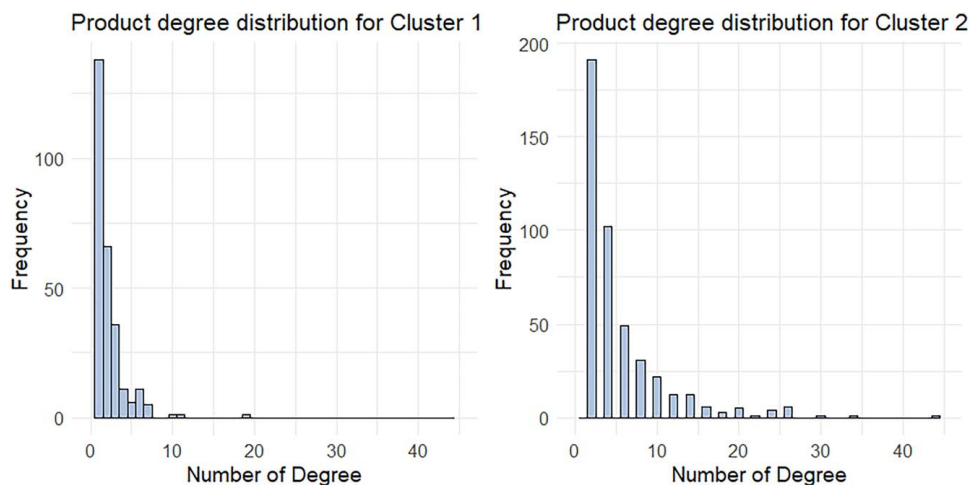
preferences between the two distinct customer clusters. Customers in the “innovation-passionate” cluster place greater emphasis on product features and are willing to pay more for superior products. In contrast, customers in the “price-sensitive” cluster exhibit a stronger preference for upright vacuum cleaners with more traditional designs. Furthermore, product features exert a more significant influence during the consideration stage, while their impact diminishes during the choice stage.

We also run the baseline network model that does not involve market segmentation. The estimated parameters in each stage are recorded in Table 5. The single bipartite network uses data that composes 871 customers and 528 products. We notice that the obtained coefficient of market distribution ( $-1.0540$ ) in the single network at the consideration stage is negative, meaning that the market distribution is more skewed, and some products are more frequently considered than the rest of the products in the market. On the contrary, with market-segment-based network modeling, the products are more evenly considered by customers in customer cluster 1. This finding suggests that, when viewed holistically, the

vacuum cleaner market is dominated by a few key players. The market-segmentation-based models, on the other hand, enable us stratify two distinct sub-markets—one that is highly competitive and another that remains monopolistic. For preference toward vacuum cleaner type, the single model shows that both upright and robotic vacuum cleaners are preferred compared to other types of vacuum cleaners but it can't differentiate the specific product attributes that are influential for each type. The market-segment-based models overcome this difficulty and we can observe the differences in preferences for upright and robotic vacuum cleaners for different customer clusters in Table 4. This observation also applies to the investigated product features. Although the baseline model exhibits similar trends for feature importance as the market-segmented model, it fails to differentiate varying preferences for each product feature among different customer clusters.

**3.4.3 Step 3: Model Prediction and Validation.** In addition to interpreting the important features in different customer segments,





**Fig. 9 Degree distribution of products in consideration networks for customer clusters. Customer cluster 2 exhibits a more pronounced long-tail effect, with a few notably popular products. In contrast, products in customer cluster 1 generally have degrees that are closer together, indicating a more evenly distributed popularity.**

we also validate the market-based-segmentation model performance by investigating the predictability of the model. The prediction accuracy of customers' considerations and choices serves as a validation metric for the effectiveness of the market segmentation compared to a benchmark model and could potentially be generalized to network-based choice modeling predictions.

The predictions for testing customers' considerations and choices are based on the ERGM. Initially, we consider any potential interactions between these testing customers and all products as missing links [38,39], as shown in Fig. 10. This step is crucial during the model estimation process, where we only rely on the non-missing links to understand the pattern in our network, excluding these "missing" interactions from our calculations to avoid biasing our results. After estimating the model parameters, we employ a simulation process to generate networks incorporating the same nodes (customers and products) following the ERGM formula in Eq. (1). The simulation uses Gibbs sampling, which is embedded in the Statnet package for ERGM [18]. In the simulation process, we treat the connections between training customers and their considerations or choices as "observed links," which remain unchanged. Due to the stochastic nature of the simulation, we generate 500 networks to capture a wide range of possible outcomes. By averaging the results across these simulations, we can estimate the likelihood of any interaction between testing customers and

products within our network. The outcomes of this simulation serve as predictions for the customers' considerations and choices. This approach allows us to infer the customer-product interactions, offering a robust method for making reasonable predictions for unknown customers.

In the consideration stage, we identify the products most likely to be considered by each customer based on the probabilities of the simulated links. We rank the potential products according to their likelihood of being considered and identify the top  $N$ . These products are then compared to the actual products considered, as reported by the testing customers. We calculate a hit rate by determining the percentage of actual products covered by the top  $N$  product sets. This hit rate allows us to evaluate the overall accuracy of the model. Figure 11 illustrates the hit rate of product consideration prediction for different models, with both random prediction and network without segmentation serving as benchmark models. The area under the curve has also been calculated as a quantitative indicator of the model's overall performance; an area equal to 1 represents a perfect and accurate prediction. As shown in the figure, the network with segmentation improves the overall predicted hit rate by 21.2%.

In the choice stage, we also simulate the network, but this time we restrict each customer to form a purchase link only within their actual consideration set. In other words, we predict customers' choices conditional on their consideration sets. By calculating the probability of purchase link existence using 500 simulated networks, we identify the product with the highest probability as the customer's choice within their consideration set. In the original survey data, there are 28.7% of respondents who reported only one product in their consideration set, and that product was also the one they ultimately purchased. As such, predicting their choices based on their consideration set is not meaningful in these cases. Therefore, when calculating prediction accuracy, we remove customers who only consider one product. The remaining customers have two to seven products in their consideration sets. As a baseline, random guessing predicts 36.67% of customer choices correctly, and the single network model without segmentation yields 50% correct predictions. The market segmentation model further improves prediction accuracy to 56.32%.

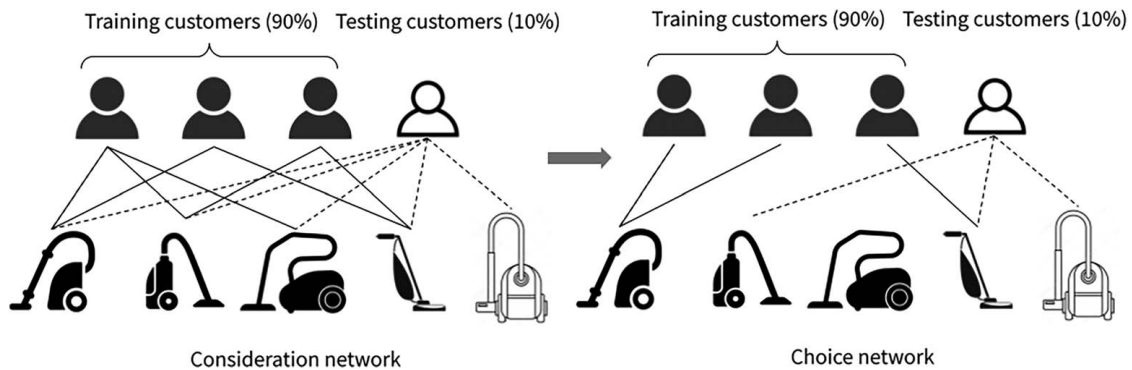
In summary, network-based models have demonstrated their predictive capabilities in forecasting customer considerations and choices, given the ERGM and the inherent network structures. Additionally, the predictive power of the market-segmentation-based model surpasses that of the single network model without segmentation, indicating that market segments effectively capture the heterogeneity in customer preferences. There are still limitations

**Table 5 ERGM results of two-stage modeling results for a single network model (without market segmentation)**

| Model terms            | Stage 1 (consideration) | Stage 2 (choice) |
|------------------------|-------------------------|------------------|
| Edges                  | -5.8230 ***             | -0.2863          |
| Market distribution    | -1.0540 ***             | -0.5529 ***      |
| Product shared partner | -0.3418 ***             |                  |
| Upright vacuum         | 0.3173 ***              | 0.4351 ***       |
| Robotic vacuum         | 0.3340 **               | -0.0622          |
| Price                  | 0.0002                  | 0.0001           |
| Filter: HEPA           | -0.2112 **              | 0.0892           |
| Capacity               | 0.0169 ***              | 0.0177           |
| Bagless                | 0.4079 ***              | 0.1977           |
| Suction power          | 0.1341 ***              | -0.0941          |
| Surface match          | 0.2188 ***              |                  |
| Model fit: AIC         | 23,288                  | 4023             |
| Model fit: BIC         | 23,420                  | 4132             |

Note:  $p < 0.1$ ; \*  $p < 0.05$ ; \*\*  $p < 0.01$ ; \*\*\*  $p < 0.001$ .





**Fig. 10** Illustration of missing links in consideration and choice networks (dashed lines represent missing links; within the choice network, missing links are limited to customers' consideration set)

to the predictive power of network-based models, which are addressed in the discussion section of our study.

## 4 Discussion

The prior section highlights the efficacy of our market-segmentation-based network modeling approach. Next, we delve into its implications on product design, while also addressing limitations and suggesting potential improvements.

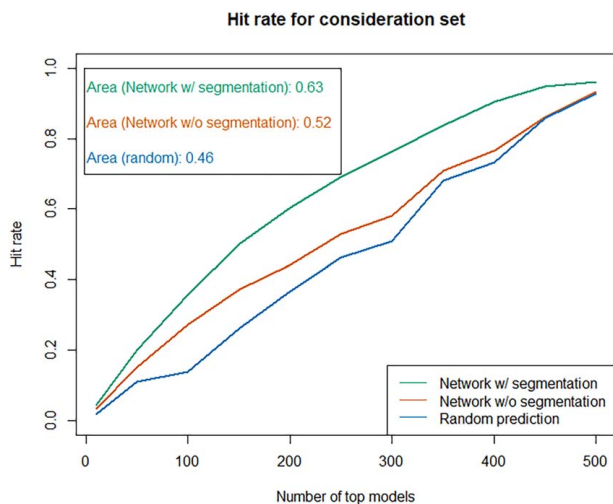
**4.1 Implication on Product Design.** The market-segmentation-based model proposed in this study enhances the applicability of network-based methods in customer preference modeling, particularly for **diverse and mixed product markets**. This approach stratified the market into finer layers and allowed product designers to distinguish customer groups with different needs, and thus customize their designs. In the vacuum cleaner market, we observed that customers in the price-sensitive segment primarily purchase upright vacuum cleaners. These customers tend to prioritize cost over advanced features like capacity, bagless design, and suction power, unlike the innovation-passionate group, which values these attributes highly. For the price-sensitive group, product designers can focus on creating cost-effective, upright vacuum cleaners that meet the essential needs of this segment without overburdening the product with unnecessary features. This might involve designing a line of vacuum cleaners

that, while being budget-friendly, still offer sufficient suction power and durability to satisfy the basic requirements of these consumers. For the innovation-passionate group, product designers may experiment with advanced features like enhanced suction technology, ergonomic, and sustainable designs, smart connectivity for home automation integration, and more efficient, bagless models. This segment's willingness to pay a premium for high-end features justifies the investment in research and development to introduce breakthrough innovations. Designers can leverage cutting-edge materials and technology to meet these consumers' expectations, ensuring that the products not only excel in functionality but also stand out in a crowded market through unique selling points. Recognizing these distinctions allows product designers to allocate resources more efficiently, tailoring designs to different market needs and catering to specific customer preferences.

This approach not only caters to diverse customer needs but also offers a more comprehensive understanding of the different driving factors in customers' consideration and choice stages. By examining various factors that play critical roles at different stages, product designers can develop tailored strategies accordingly. For instance, if products are often considered by customers but rarely chosen, it is essential to identify the factors that influence customers' choices during the evaluation of their consideration set. By addressing these factors, businesses can better cater to customers and improve their products' chances of being chosen.

**4.2 Limitations.** Building on the implication of our research, it is essential to acknowledge some limitations that may impact the findings of this study. First, our survey data is limited in size, which may impact the robustness of our results. A larger dataset that captures a more diverse range of customer preferences and product markets would provide a more comprehensive understanding of the relationships identified in our study. Second, while the network-based model demonstrates effectiveness in capturing the interdependence of the customer decision-making process and outperforms traditional choice models, its predictive capability has inherent limitations. The linear nature of the model restricts its ability to make highly accurate predictions, particularly when compared to more advanced machine learning models that can accommodate complex non-linear relationships. Despite these limitations, our research offers a valuable foundation for future studies to build upon. Researchers may consider expanding the dataset size and exploring alternative modeling approaches to enhance the predictive power of network-based customer preference models.

**4.3 Comparison With Graph Neural Network.** In our study, we recognized the limitations of using ERGMs due to their linear form and fixed structure, leading us to consider advanced



**Fig. 11** Comparison of hit rate performance for three different models: (1) random selection, (2) single network without market segmentation, and (3) market-segmented model

alternatives like graph neural networks (GNNs) for better prediction accuracy. Our preliminary exploration [40] shows GNNs' potential to enhance predictions of customers' considerations and choices through their complex, non-linear models. GNNs enable us to categorize products within the product co-consideration network and predict links (customer-product relationships) in bipartite networks, considering the interdependencies among network entities. Specifically, when dealing with two types of nodes (customers and products) and various link types (consideration and choice), it becomes essential to apply the concept of a heterogeneous graph. However, the enhanced prediction accuracy provided by GNNs entails a tradeoff with interpretability, a crucial aspect of our research aimed at understanding customer preferences. Despite GNNs' predictive advantages, we prioritize methods that offer a balance between accuracy and insight into customer decision-making, thereby offering actionable insights for product design.

## 5 Conclusion

In this research, we propose a novel approach for identifying heterogeneous customer preferences in the two-stage consideration-then-choice decision-making process. Our analysis of real-customer survey data for the vacuum cleaner market indicates that product attributes play more important roles in the consideration stage compared to the choice stage. Additionally, the same product attributes could have varying effects on different market segments (innovation-passionate customers versus price-sensitive customers in our case study). We further validate the market-segmentation-based network models by comparing their predictive power to a benchmark model. The results show that network-based models with market segmentation not only offer a more practical interpretation of customer preferences by reflecting diverse customer tastes compared to a single network model, but also provide more accurate estimations of customers' considerations and choices.

To the authors' knowledge, this paper is the first study to investigate customer segmentation in network-based customer preference modeling to gain insights into complex heterogeneous customer preferences. The contributions of the research methodology can be outlined as follows: First, traditional study segments a market by using either specific aspects of customer attributes or all available customer attributes without examining their relevance and association with preferences. In our study, we propose a data-driven approach that selects meaningful customer attributes reflecting demographic attributes, usage contexts, and personal viewpoints to identify market segmentation. Second, our model introduces a network-based approach across various stages of the customer's decision-making process within a diverse and mixed product market. This represents the first application of network-based customer preference modeling beyond its initial test dataset (i.e., the car market). Enabled by market segmentation, our model can capture distinct market patterns and heterogeneous customer preferences within two sub-networks. Importantly, it also uncovers unique factors influencing customers during the consideration and choice stages of their decision-making process. Furthermore, this paper presents a new validation method for network-based models based on network predictions. This method preserves a portion of network links as missing edges and predicts the network by simulating new networks. This validation approach enables the comparison of the performance of network-based models with different data and structures. In our study, we demonstrate the effectiveness of our proposed model by comparing it with a benchmark model using this validation method. In summary, our method presents a systematic market-segmentation-based network approach for investigating customer preferences in their consideration-then-choice decision-making behavior. This research lays the groundwork for future studies exploring more comprehensive network-based methods for examining customer preferences across different product markets.

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## Conflict of Interest

There are no conflicts of interest.

## Data Availability Statement

The data and information that support the findings of this article are freely available online.<sup>3</sup>

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